# Exercise 13

# On equation of a plane in space

From the school book | Understand | Apply | Higher Order Thinking Skills

## 1 Choose the correct answer from those given :

( 1 ) The equation of the plane $x$ y is .....

(a) $x = 0$ (b) $y = 0$ (c) $z = 0$ (d) $x + y = 0$

( 2 ) The normal direction vector to the plane : $-2x - 4y + z - 15 = 0$ is .....

(a) (2 , 4 , 1) (b) (2 , 4 , 5) (c) (2 , – 4 , – 1) (d) (– 2 , – 4 , 1)

( 3 ) If (5 , –1 , 2). $\vec{r} = 0$ is the vector form of the equation of the plane passes through the origin point, then the general form is .....

(a) $5x - y + 2z = 6$ (b) $5x - y + 2z = 0$

(c) $y - 5x + 2z = 8$ (d) $2x - y + 5z = 0$

( 4 ) If $3x - 5y - 4z - 27 = 0$ is the general from of the equation of the plane, then the vector form of its equation is .....

(a) (3 , – 5 , – 4) . $\vec{r} = 27$ (b) (3 , – 5 , – 4) . $\vec{r} = -27$

(c) (5 , 4 , 27) . $\vec{r} = 3$ (d) (3 , 4 , 5) . $\vec{r} = 27$

( 5 ) The equation of the plane passing through point (1 , –2 , 5) and the vector (2 , 1 , 3) is perpendicular to it is .....

(a) $2x + y + 3z = 1$ (b) $2x + y + 3z = 15$

(c) $x - 2y + 5z = 15$ (d) $x + y + z = 4$

( 6 ) The point (2 , –1 , 3) lies on the plane .....

(a) $x + y - z = 6$ (b) $2x - 3y + z = -10$

(c) $3x - 2y + 4z = 20$ (d) $x - 2y + 5z = 4$

( 7 ) Which of the following points belongs to the plane $2x + 3y - z = 5$ .....

(a) (1 , 1 , 1) (b) (1 , 2 , 0) (c) (0 , 2 , 1) (d) (3 , 2 , –1)

( 8 ) If A (1 , 0 , 0) and B (0 , 1 , 1) lie on the plane $kx + y + mz + 2 = 0$, then $k + m$ = .....

(a) –7 (b) –5 (c) –3 (d) –1

**( 9 )** The direction cosines of the normal to the plane : $3x + 4y + z = 1$ is .....

(a) $\pm\left(\frac{3}{5}, \frac{4}{5}, \frac{1}{5}\right)$ (b) $\pm(3, 4, 1)$

(c) $\pm\left(\frac{3}{\sqrt{26}}, \frac{4}{\sqrt{26}}, \frac{1}{\sqrt{26}}\right)$ (d) $\pm\left(\frac{1}{\sqrt{26}}, \frac{4}{\sqrt{26}}, \frac{3}{\sqrt{26}}\right)$

**(10)** The equation of the plane which parallel to the plane yz and passes through the point $(1, 3, 5)$ is .....

(a) $x = 1$ (b) $y = 3$ (c) $z = 5$ (d) $y + z = 8$

**(11)** The equation of the plane passing through the origin point and parallel to the plane $3x - 4y + 5z - 6 = 0$ is .....

(a) $3x - 4y + 5z = 0$ (b) $5x - 4y + 3z = 0$

(c) $3x - 4y + 5z = 2$ (d) $5x - 4y + 3z = 4$

**(12)** $2y + 3 = 0$ is an equation of a plane .....

(a) parallels the plane (y z) (b) parallels the plane (x z)

(c) parallels the plane (x y) (d) otherwise.

**(13)** ***(2nd Session 2021)*** The general form of the equation of the plane which passing through the point $(-2, 2, -1)$ and parallel to the plane whose equation : $(2, 3, -5) \cdot \vec{r} = 1$ is .....

(a) $2x + 3y - 5z = -7$ (b) $2x + 2y - z = 1$

(c) $2x - 3y + 5z = -7$ (d) $2x + 3y - 5z = 7$

**(14)** The plane is determined by .....

(a) three distinct non collinear points. (b) a straight line and a point outside it.

(c) two intersecting straight lines. (d) all the previous.

**2** Find the general form of the equation of the plane passing through the origin point and the vector $\vec{n} = \hat{i} + 2\hat{j} - 3\hat{k}$ is normal to the plane.

**3** ***(2nd Session 2019)*** Find all the different forms of the equation of the plane passing through the point $(-3, 4, 2)$ and the vector $\vec{n}$ $(1, -1, 3)$ is normal to the plane.

**4** ***(2nd Session 2018)*** Find all the different forms of the equation of the plane passing through point $(1, -1, 4)$ and its normal vector is $\vec{n} = (2, -3, 4)$, then show the following : **( 1 )** Is point $(2, 2, 1)$ lying on the plane ?

**( 2 )** Is the vector $\vec{u} = (3, -5, -2)$ parallel to the plane ?

**5** Find all the difference forms of the equation of the plane that intercepts 2, 4, 5 from the coordinate axes x, y, z respectively.

**6** ***(1st Session 2019)*** Find all the different forms of the equation of the plane passing through points $(1, 0, 0)$, $(0, 2, 0)$ and $(0, 0, 3)$

**7** Find all the different forms of the equation of the plane passing through the three points :

( 1 ) A (2 , – 1 , 0) , B (– 1 , 3 , 4) , C (3 , 0 , 2)

( 2 ) (3 , 0 , 4) , (2 , 1 , 1) , the origin point.

**8** Find three points in space belonging to each of the following planes :

( 1 ) $x = 3$ ( 2 ) $y = -2$

( 3 ) $x + 3y = 5$ ( 4 ) $2x - y + 3z = 4$

**9** Determine position of each of the points A (3 , – 2 , 5) , B (– 4 , 1 , 0) , C (6 , 5 , – 1) , D (– 3 , – 5 , 2) , E (4 , – 7 , 10) with respect to the plane $2x - 3y + 4z - 5 = 0$

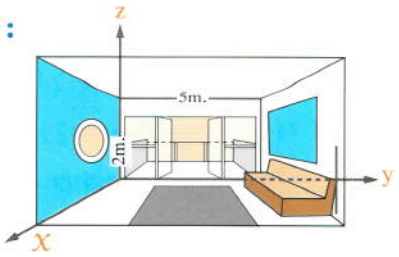
**10** Find the intercepted parts made by the plane $2x + 3y - z = 6$ with the coordinate axes.

**11** **In the opposite figure, find the equation for each of :**

( 1 ) The floor plane.

( 2 ) The cieling plane.

( 3 ) The lateral walls planes.




**12** **Choose the correct answer from those given :**

( 1 ) *(1st Session 2017)* The equation of the plane which passes through the point (1 , 2 , 3) and parallel to the coordinate axes $x$ , y is .....

(a) $x + y = 3$ (b) $z = 3$ (c) $x = 1$ (d) $y = 2$

( 2 ) The equation of the plane passing through points (2 , 3 , 5) , (–1 , 3 , 1) , (4 , 3 , – 2) is .....

(a) $x + y - z = 0$ (b) $x = -1$ (c) $y = 3$ (d) $z = -2$

( 3 ) The equation of the plane which parallel to the $x$-axis is .....

(a) $a x + c z + d = 0$ (b) $b y + c z + d = 0$

(c) $a x + b y + d = 0$ (d) $a x + b y + c z + d = 0$

( 4 ) In space of three dimensions the equation $3y + 4z = 0$ represents .....

(a) a plane contains z-axis (b) a plane contains $x$-axis

(c) a plane contains y-axis

(d) a straight line its direction ratios (0 , 3 , 4)

( 5 ) $4x + 5y - 9 = 0$ is the equation of a plane .....

(a) parallel to $x$-axis. (b) parallel to z-axis.

(c) contains the y-axis. (d) contains parallel to z-axis.

**( 6 )** If the straight line $x = 3y = az$ is parallel to the plane $x + 3y + 2z + 4 = 0$ , then $a =$ .....

(a) $-3$ (b) $3$ (c) $-1$ (d) $1$

**( 7 )** *(2nd Session 2023)* The straight line whose equation : $\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-1}{4}$ intersects the plane whose equation $2x + y - z = 5$ at the point .....

(a) $(2, 3, 9)$ (b) $(5, 4, 9)$ (c) $(5, -2, 9)$ (d) $(5, 4, 5)$

**( 8 )** The two points $A(-2, 3, 5)$ , $B(4, -7, -2)$ , the plane $P : 2x - 3y + z - 7 = 0$ , then A , B .....

(a) belong to the plane P

(b) lies on the same side of the plane P

(c) lies on two different sides of the plane P

(d) one of the two points lies on the plane P

**( 9 )** The plane $3x - 2y + 4z = 12$ intercepts from x-axis a part of length ..... length unit.

(a) 3 (b) $-4$ (c) 4 (d) 6

**(10)** If the intercepted parts from the coordinate axes by the plane $x + 5y - 6z = 30$ are a , b , c, then $a + b + c =$ .....

(a) 0 (b) 30 (c) 31 (d) 41

**(11)** If the plane $\frac{x}{4} + \frac{y}{2} + \frac{z}{2} = 1$ cuts the coordinate axes at points A, B, C , then the area of the triangle A B C = .....

(a) 12 (b) 10 (c) 6 (d) 4

**(12)** If a plane cuts the coordinate axes at A , B , C where the intersection point of the medians of $\Delta$ ABC is $(3, 3, 3)$, then the equation of the plane is .....

(a) $3x + 3y + 3z = 1$ (b) $3x + 3y + 3z = 3$

(c) $\frac{x}{3} + \frac{y}{3} + \frac{z}{3} = 1$ (d) $\frac{x}{3} + \frac{y}{3} + \frac{z}{3} = 3$

**(13)** The plane equation which intercepts from the positive x-axis and z-axis parts of length 2 and 4 length units and from the negative y-axis part of length 3 length units is .....

(a) $2x - 3y + 4z = 1$ (b) $6x - 4y + 3z = 6$

(c) $6x + 4y + 3z = 12$ (d) $6x - 4y + 3z = 12$

**(14)** The plane $20x + 15y + 12z = 60$ intersects the coordinate axes $x$ , $y$ , $z$ at the points A , B , C respectively then the volume of the solid ABCO where O is the origin equals ..... cubic unit.

(a) 30 (b) 60 (c) 90 (d) 10

**(15)** The equation of the plane which is perpendicular to the vector : $2\hat{i} + \hat{j} + 3\hat{k}$ and passes through the point (1 , 2 , 3) is .....

(a) $2x + y + 3z = 0$ (b) $2x + y + 3z = 13$

(c) $2x + y + 3z = 14$ (d) $2x + y + 3z = 7$

**(16)** The plane whose equation $2x - y = 0$ is perpendicular to the plane .....

(a) $xy$ (b) $yz$ (c) $xz$ (d) $y - 2x = 0$

**(17)** The plane contains the origin point and the vector (0 , 1 , – 2) is perpendicular to it is .....

(a) $x = 2y$ (b) $y = 2z$ (c) $z = 2x$ (d) $y = 1$

**(18)** The equation of the straight line passing through the point (1 , 1 , 1) and is perpendicular to the plane $2x + 3y + z + 5 = 0$ is .....

(a) $\frac{x-1}{-1} = \frac{y-1}{1} = \frac{z-1}{1}$ (b) $\frac{x-1}{2} = \frac{y-1}{3} = \frac{z-1}{1}$

(c) $\frac{x-1}{3} = \frac{y-1}{3} = \frac{z-1}{1}$ (b) $\frac{x-1}{1} = \frac{y-1}{3} = \frac{z-1}{2}$

**(19)** The equation of the straight line passing through the origin perpendicular to the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ is .....

(a) $\vec{r} = (0 , 0 , 0) + t(a , b , c)$ (b) $\vec{r} = (0 , 0 , 0) + t(\frac{1}{a} , \frac{1}{b} , \frac{1}{c})$

(c) $\vec{r} = (a , b , c) + t(0 , 0 , 0)$ (d) $\vec{r} = (a , b , c) + t(1 , 1 , 1)$

**(20)** The equation of the plane contains the z-axis and passes through (1 , –1 , 3) is .....

(a) $3x + y = 6$ (b) $3x + y = 0$ (c) $x + y = 0$ (d) $x - 4y + 3z = 0$

**(21)** The equation of the plane passes through the point (4 , 0 , 1) parallel to the plane $4x + 3y - 12z + 6 = 0$ is .....

(a) $4x + 3y - 12z = 6$ (b) $\vec{r} . (4 , 3 , -12) = 4$

(c) $4x + 3y - 12z = 0$ (d) $\frac{x}{4} + \frac{y}{3} + \frac{z}{-12} = 1$

**(22)** The straight line : $\frac{x-1}{2} = \frac{y+2}{3} = 1 - z$ and the plane $3y - z + 2x = 3$ .....

(a) are parallel. (b) the straight line lies on the plane.

(c) the straight line is perpendicular to the plane.

(d) the straight line intersects the plane but it is not perpendicular to it.

**(23)** If the straight line $\frac{x-1}{4} = \frac{y-2}{b} = \frac{z}{6}$ never intersect the plane $2x - 4y + 3z = 5$ , then b = .....

(a) $-10$ (b) $-8$ (c) $\frac{13}{2}$ (d) $\frac{19}{2}$

**(24)** If the straight line $\frac{x-2}{3} = \frac{y+1}{2} = 1 - z$ intersects the two planes $2x + 3y - z + 13 = 0$ and $3x + y + 4z = 16$ at the two points A and B , then AB = .....

(a) $\sqrt{14}$ (b) $\sqrt{28}$ (c) $2\sqrt{14}$ (d) 14

**(25)** The geometric description of set of points in space that satisfy pair of the equation $y = 3$ , $z = 1$ is a straight line parallel to the plane .....

**(1)** yz **(2)** $x$ z **(3)** $x$ y

(a) **(1)** , **(2)** (b) **(2)** , **(3)** (c) **(1)** , **(3)** (d) All the previous.

**(26)** The straight line $\frac{x-x_1}{\ell} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ lies on the plane $ax + by + cz + d = 0$ if .....

(a) $a\ell + bm + cn = 0$

(b) $ax_1 + by_1 + cz_1 + d = 0$

(c) $a\ell + bm + cn = 0$ , $ax_1 + by_1 + cz_1 + d = 0$

(d) $(\ell , m , n) \times (a , b , c) = \vec{0}$

**(27)** The two straight lines $\ell_1 : \frac{x+1}{-1} = \frac{y-2}{-1} = \frac{z-1}{3}$ , $\ell_2 : \frac{x+1}{1} = \frac{y-2}{-2} = \frac{z-1}{-1}$ lie on the plane .....

(a) $3x - 5y + z - 1 = 0$ (b) $5x - 4y + 2z - 7 = 0$

(c) $7x - 5y - z - 4 = 0$ (d) $7x + 2y + 3z = 0$

**(28)** The straight line $\frac{x+1}{1} = \frac{y+3}{3} = \frac{z-2}{-2}$ is perpendicular to the plane .....

(a) $x - 3y - 2z + 7 = 0$ (b) $2x + 6y - 4z + 3 = 0$

(c) $x + 3y - 5z = 0$ (d) $2x + 6y + 4z = 5$

**(29)** All the following points lie on the same side from the plane $\vec{r} . (3\hat{i} - 5\hat{j} + 4\hat{k}) = 5$ except .....

(a) (1 , 2 , 1) (b) (0 , –1 , 2) (c) (2 , –3 , 1) (d) (1 , 2 , 4)

**(30)** The straight line $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ is parallel to the plane .....

(a) $2x + y - 2z = 0$ (b) $x + y + z = 0$

(c) $3x + 4y + 5z = 7$ (d) $2x + 3y + 4z = 0$

**(31)** If the straight line $\ell : \frac{x-1}{3} = \frac{y+3}{2} = \frac{z-3}{-1}$ and the plane $P : x - 2y - z = 0$, then which of the following is correct ?

(a) $\ell \parallel P$ (b) $\ell \perp P$ (c) $\ell \subset P$ (d) $\ell$ cuts P

**(32)** The straight line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ with respect to the plane $x - 2y + z - 6 = 0$ .....

(a) the straight line is parallel to the plane

(b) the straight line is perpendicular to the plane

(c) the straight line lies on the plane

(d) the straight line is incline to the plane

**(33)** The measure of the angle between the straight line : $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z+3}{-2}$ and the plane : $x + y + 4 = 0$ is .....

(a) zero (b) 30° (c) 45° (d) 90°

**(34)** The point that belongs to the plane :

$\vec{r} = (-1, 0, 2) + t(0, 0, 1) + m(1, 0, -1)$ is .....

(a) $(0, 1, 2)$ (b) $(2, 1, 3)$ (c) $(3, 1, 2)$ (d) $(1, 0, 1)$

**(35)** The equation of the plane which contains the line $\vec{r} = \hat{i} + \hat{j} + t(2\hat{i} + \hat{j} + 4\hat{k})$ from the following equations is .....

(a) $\vec{r} \cdot (-\hat{i} - 2\hat{j} + \hat{k}) = 3$ (b) $\vec{r} \cdot (\hat{i} + 2\hat{j} - \hat{k}) = 6$

(c) $\vec{r} \cdot (\hat{i} + 2\hat{j} - \hat{k}) = 3$ (d) $\vec{r} \cdot (-\hat{i} - 2\hat{j} + \hat{k}) = 6$

**(36)** *(1st Session 2023)* The equation of the plane which contains the straight line whose equation : $x = y = \frac{1}{2}z$ and passes through the point $(1, 2, 3)$ is .....

(a) $\vec{r} \cdot (1, -1, -1) =$ zero (b) $x + 2y + 3z =$ zero

(c) $x - y + z =$ zero (d) $\vec{r} \cdot (1, 1, -1) =$ zero

**(37)** If the projection of the point O $(0, 0, 0)$ on the plane (P) is M $(2, 3, 1)$ then the equation of this plane is .....

(a) $2x + 3y + z - 14 = 0$ (b) $x + 3y + 2z - 14 = 0$

(c) $2x + 3y + z = 0$ (d) $x + 3y + 2z = 0$

**(38)** The image of the point $(-2\ ,3\ ,-4)$ by reflection in the $\mathcal{X}$ z-plane is .....

(a) $(-2\ ,3\ ,-4)$ (b) $(2\ ,3\ ,-4)$ (c) $(-2\ ,-3\ ,-4)$ (d) $(-2\ ,3\ ,4)$

**(39)** The two non-parallel straight lines $\vec{r}_1 = \vec{A}_1 + t_1 \vec{d}_1$ , $\vec{r}_2 = \vec{A}_2 + t_2 \vec{d}_2$ lie on the same plane if .....

(a) $\vec{A}_1 \times \vec{A}_2 = \vec{O}$ (b) $\vec{d}_1 \times \vec{d}_2 = \vec{O}$

(c) $(\vec{A}_1 - \vec{A}_2) \cdot (\vec{d}_1 \times \vec{d}_2) = O$ (d) $\vec{d}_1 \cdot \vec{d}_2 = O$

**(40)** The equation of the plane which bisects the line between the two points $(2\ ,3\ ,4)$ , $(6\ ,7\ ,8)$ may be .....

(a) $\mathcal{X} + y + z - 15 = 0$ (b) $\mathcal{X} + z - y - 15 = 0$

(c) $\mathcal{X} - y - z - 15 = 0$ (d) $\mathcal{X} + y + z + 15 = 0$

**(41)** The angle between the straight line $\vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + t(-\hat{i} + \hat{j} + \hat{k})$ and the plane $\vec{r} \cdot (3\hat{i} + 2\hat{j} - \hat{k}) = 4$ is .....

(a) $\cos^{-1}\left(\frac{2}{\sqrt{42}}\right)$ (b) $\cos^{-1}\left(\frac{-2}{\sqrt{42}}\right)$ (c) $\sin^{-1}\left(\frac{2}{\sqrt{42}}\right)$ (d) $\sin^{-1}\left(\frac{-2}{\sqrt{42}}\right)$

**(42)** The point A is the image of the point N $(1\ ,2\ ,3)$ by reflection on the plane $\mathcal{X}$ y and B is the image of A by rotation about the origin with an angle of measure 180° and C is the image of B by translation 5 units in positive direction of y-axis , then C = .....

(a) $(1\ ,7\ ,-3)$ (b) $(-1\ ,7\ ,-3)$ (c) $(-1\ ,-2\ ,8)$ (d) $(-1\ ,3\ ,3)$

**(43)** If the point B $(2\ ,1\ ,3)$ is the projection of the point A $(\mathcal{X}_1\ , y_1\ ,6)$ on the plane $2\mathcal{X} + y + z = 8$ , then $\mathcal{X}_1 \times y_1$ = .....

(a) 10 (b) 16 (c) 32 (d) 42

**(44)** If the distance between point (A) and the plane (P) equals 16 length units and the straight line passing through (A) intersects the plane (P) at point (B) and makes with the plane (P) an angle of measure $\theta$ such that $\tan\theta = \frac{8}{15}$ , then length of the projection of $\overline{AB}$ on the plane = ..... length units

(a) 30 (b) 15 (c) 16 (d) 8

**(45)** If measure of the angle between the straight line : $\frac{\mathcal{X} - 5}{2} = \frac{y}{-1} = \frac{z + 4}{a}$ and the plane $3\mathcal{X} + 2y + z = 5$ equals 30° , then a = ..... where $a \in \mathbb{Z}$

(a) $-3$ (b) 2 (c) 3 (d) $-4$

**(46)** If the point $(4 , -1 , 3)$ is the image of the point $(2 , 1 , 1)$ by reflection on a plane , then equation of that plane is .....

(a) $\mathcal{X} - y + z = 5$ (b) $4\mathcal{X} - y + 3z = 10$

(c) $2\mathcal{X} + y + z = 5$ (d) $\mathcal{X} - 2y + z = 5$

**(47)** If $\theta$ is the measure of the angle that the plane $2\mathcal{X} - 3y + 6z = 11$ makes with the positive direction of $\mathcal{X}$-axis , then $\sin\theta =$ .....

(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{2}{7}$ (c) $\frac{\sqrt{2}}{3}$ (d) 1

**(48)** ***(Trial 2021)*** If the straight line : $\frac{\mathcal{X}-2}{3} = \frac{y+1}{-4} = \frac{z+3}{5}$ makes angles of measures $\ell , m , n$ with the $\mathcal{X}$ y-plane , y z-plane and $\mathcal{X}$ z-plane respectively , then

$\sin^2\ell + \sin^2 m + \sin^2 n =$ .....

(a) 1 (b) 2 (c) $\frac{3}{2}$ (d) $\sqrt{3}$

**(49)** ***(Trial 2021)*** If the plane $bc\mathcal{X} + acy + abz = abc$

intercepts the coordinate axes at the point K , N , M respectively and the plane : $bc\mathcal{X} + acy - abz = -abc$ intercepts the coordinate axes at the points $\grave{K}$ , $\grave{N}$ , M respectively , then the pyramid M K N $\grave{K}$ $\grave{N}$ is .....

where a , b , c are positive real numbers , $a \neq b$

(a) a right pyramid (b) a regular quadrilateral pyramid

(c) a right triangular pyramid (d) a regular triangular pyramid

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**13** If a plane intersects the coordinate axes at the points A , B , C and the point $(m , n , k)$ is the point of intersecting of the medians of triangle ABC

Prove that the equation of the plane is $\frac{\mathcal{X}}{m} + \frac{y}{n} + \frac{z}{k} = 3$

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**14** ***(1st Session 2017)*** If the plane $3\mathcal{X} + 2y + 4z = 12$ intersects the coordinate axes $\mathcal{X}$ , y , z at the points A , B , C respectively , find the area of the triangle ABC « $3\sqrt{29}$ square unit »

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**15** Find the equation of the straight line passing through the point $(\mathcal{X}_1 , y_1 , z_1)$ and is perpendicular to the plane $a\mathcal{X} + by + cz + d = 0$

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**16** **Find the equation of the plane :**

**( 1 )** That parallels to the plane $\mathcal{X} - 2y + 2z = 1$ and passes through the point $(1 , -1 , 2)$

**( 2 )** That passes through the two points A $(1 , 2 , -3)$ , B $(2 , 3 , -4)$ and is perpendicular to the plane $\mathcal{X} + y + z + 1 = 0$

( 3 ) That passes through the point $(1 , 2 , 3)$ and parallels to the two vectors $(2 , 1 , -1)$ , $(3 , 6 , -2)$

( 4 ) That passes through the two points $(1 , -2 , 5)$ , $(2 , 2 , 3)$ and parallels the vector $\overrightarrow{A} = (3 , -1 , 0)$

( 5 ) That passes through the point $(4 , -1 , 2)$ and contains the $x$-axis.

( 6 ) That the projection of the origin point on it is the point $(2 , 3 , -5)$

( 7 ) Which intercepts the coordinate axes $x$ , y , z the parts $2 , -3 , 5$ respectively.

( 8 ) That passes through the point $(3 , 5 , -7)$ and cuts equal parts from the three coordinate axes.

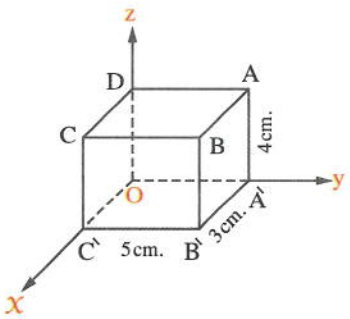
( 9 ) That passes through the two points A $(3 , 5 , 1)$ , B $(7 , 7 , 8)$ and cuts equal parts from the two coordinate axes $x$ , y

(10) That perpendicular to the plane $2x - 2y + 4z - 5 = 0$ and cuts the two axes $x$ and y at $(-2 , 0 , 0)$ , $\left(0 , \frac{2}{3} , 0\right)$

(11) That intercepts from the negative part of $x$-axis a part equals 3 units and from the positive part of z-axis 2 units and passes through the point $(6 , -10 , 1)$

(12) That cuts $x$-axis at the point $(5 , 0 , 0)$ and y-axis at the point $(0 , -7 , 0)$ and passes through the point $(1 , 1 , 2)$

**17** **By using the opposite figure find the equation of each of the following :**




( 1 ) The equation of $\overleftrightarrow{AB}$

( 2 ) The equation of $\overleftrightarrow{BC}$

( 3 ) The equation of $\overleftrightarrow{BB'}$

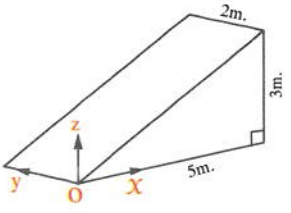
( 4 ) The equation of the plane ABCD

( 5 ) The equation of the plane $A'B'C'O$

( 6 ) The equation of $\overleftrightarrow{AC'}$

( 7 ) The equation of the plane $ABC'O$

**18** Using the opposite figure find equation of the inclined plane and also equation of the line of greatest slope passing through the origin point.



**19** Prove that the two straight lines $\vec{r}_1 = (3\hat{i} + \hat{j} - \hat{k}) + t_1(\hat{i} + 2\hat{j} + 3\hat{k})$

$\vec{r}_2 = (2\hat{i} + 5\hat{j}) + t_2(\hat{i} - \hat{j} + \hat{k})$

are intersecting , then find the equation of the plane containing them.

**20** Prove that the two straight lines $L_1 : 2x = 3y = 4z$ , $L_2 : 3x = 2y = 5z$ are intersecting , then find the equation of the plane containing the two straight lines.

**21** Prove that the two straight lines

$L_1 : \vec{r} = (1 , -2 , 2) + t(4 , 1 , 3)$ ,

$L_2 : \frac{x-3}{8} = \frac{y-4}{2} = \frac{z+5}{6}$

are parallel then determine the general equation of the plane contains these two lines.

**22** *(1st Session 2017)* Find the equation of the plane which contains the straight line $L_1 : \vec{r} = (0 , 3 , -5) + t_1(6 , -2 , -1)$ and parallel to the straight line $L_2 : \vec{r} = (1 , 7 , -4) + t_2(1 , -3 , 3)$

**23** Find the equation of the plane which contains the straight line : $\frac{x+1}{-3} = \frac{y-3}{2} = \frac{z+2}{1}$ and the point $(0 , 7 , -7)$

**24** Find the equation of the plane which contains the straight line : $x = \frac{y-3}{2} = \frac{z-5}{3}$ and perpendicular to the plane $2x + 7y - 3z = 1$

**25** Find the equation of the plane passing through the origin point and is perpendicular to each of the two planes $x + 2y - z = 1$ , $3x - 4y + z = 5$

**26** Find the equation of the plane passing through the point $(2 , 1 , 4)$ and satisfies one of the following condition :

( 1 ) Parallel to the plane $2x + 3y + 5z = 1$

( 2 ) Perpendicular to the straight line passing through the two point $(3 , 2 , 5) , (1 , 6 , 4)$

( 3 ) Perpendicular to each of the planes $7x + y + 2z = 6$ , $3x + 5y - 6z = 8$

**27** Write the general equation of the plane $x = 2 + 2t_1 - 4t_2$ , $y = 4 - t_2$ , $z = 2 + 3t_1$

**28** Find the cartesian equation of the plane

$(x , y , z) = (2 , 3 , 5) + t_1(-1 , 3 , 4) + t_2(6 , 1 , -2)$ where $t_1 , t_2$ are parameters.

**29** Prove that the straight line $\frac{x-3}{5} = \frac{y+1}{5} = \frac{z+4}{7}$ lies on the plane $3x + 4y - 5z = 25$

**30** Prove that point A $(2, 3, 1)$, the straight L : $\vec{r} = (3\hat{i} + \hat{j} + 3\hat{k}) + t(\hat{i} - 2\hat{j} + 2\hat{k})$ are lying on the plane whose equation is $\vec{r} \cdot (2\hat{i} - \hat{k}) = 3$

**31** Prove that the straight line $\vec{r} = \hat{k} + t(2\hat{i} + 3\hat{j} + 4\hat{k})$ is perpendicular to the plane $x + \frac{3}{2}y + 2z = 5$

**32** Find the measure of the angle between the straight line $\vec{r} = (\hat{i} + 2\hat{j} - \hat{k}) + t(\hat{i} - \hat{j} + \hat{k})$ and the normal to the plane $\vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) = 4$ « 19° 28` »

**33** Find the measure of the angle between the straight line $\frac{x+1}{3} = \frac{y-1}{2} = \frac{z-2}{4}$ and the plane $2x + y - 3z + 4 = 0$ « 11° 27` »

**34** If the straight line $\frac{x-3}{2} = \frac{y+3}{-1} = \frac{z}{3}$ cuts the plane $3x + 2y + z - 8 = 0$ at a point find the measure of the angle of inclination of the straight line on the plane. « 30° »

**35** If the measure of the smaller angle between the straight line : $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z+3}{-2}$ and the plane $mx + y + 4 = 0$ equals 45° find the value of m « 1 or 7 »

**36** If $\theta$ is the measure of the smaller angle between the straight line $\frac{x+3}{k} = \frac{y-2}{-2} = \frac{z+1}{1}$ and the plane $x + 2y - 3z + 5 = 0$ and $\cos\theta = \frac{5\sqrt{70}}{42}$, find the value of k « $\frac{1}{4}$ or 2 »

**37** Find the point of intersecting of the straight line $x = y = z$ and the plane $x + 2y + 3z = 12$ « (2 , 2 , 2) »

**38** Find the coordinates of the point of intersection of the straight line $\vec{r} = (2, -1, 2) + t(3, 4, 2)$ with the plane $\vec{r} \cdot (1, -1, 1) = 5$ « (2 , – 1 , 2) »

**39** Find the coordinates of the point of intersection of the straight line $\vec{r} = \hat{k} + t(2\hat{i} + \hat{j} + \hat{k})$ with the plane $\vec{r} \cdot \hat{i} = 4$ « (4 , 2 , 3) »

**40** Find the projection of the point $(1, 2, 3)$ on the plane $x + 2y + 4z = 59$ « (3 , 6 , 11) »

**41** Find the point A that lie on the plane $2x + y - 2z = 1$ and at a distance as small as possible from the point B $(1, 0, -1)$

« $\left(\frac{1}{3}, \frac{-1}{3}, \frac{-1}{3}\right)$ »

**42** Find the image of the point A $(3, -8, -7)$ by reflection in the plane : $x - 3y + z + 2 = 0$

« $(-1, 4, -11)$ »

## Higher skills

**43** Find the coordinates of the point of intersection of the straight line passing through the two points $(3, -4, -5)$, $(2, -3, 1)$ with plane passing through points $(2, 2, 1)$, $(3, 0, 1)$, $(4, -1, 0)$

« $(1, -2, 7)$ »

**44** Find the equation of the projection of the straight line $\vec{r} = (1, 2, -1) + t(0, -1, 2)$ on the plane $2x - y + z = 5$

**45** Find the point of intersection of the planes :

$2x + y - z = -1$ , $x + y + z - 2 = 0$ , $3x - y - z = 6$

« $\left(2, \frac{-5}{2}, \frac{5}{2}\right)$ »